Problem 5

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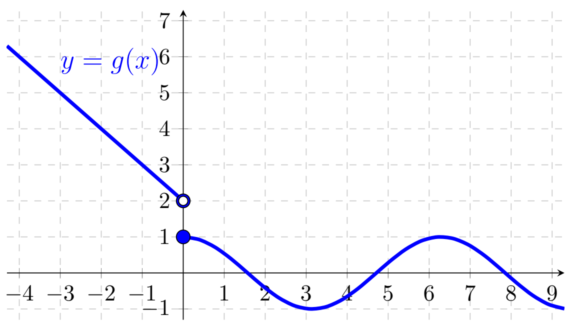
Problem 1 *Define* $g(x) = \begin{cases} |x-2| & \text{if } x < 0 \\ \cos(x) & \text{if } x \geq 0 \end{cases}$

(a) *Sketch a graph of* g

Explanation.



(b) *Find the domain and range of* g

Explanation. Domain: $(-\infty, \infty)$, Range: $[-1, 1] \cup (2, \infty)$

(c) *Find the values of* $g(\pi)$ *and* $g(-\pi)$

Explanation. $g(\pi) = \cos(\pi) = -1$ and $g(-\pi) = |-\pi - 2| = \pi + 2$